

- 28** An electron of mass m and charge e is accelerated from rest through an electric field of potential difference V .

What is the frequency of a photon whose wavelength is equal to the de Broglie wavelength of this electron?

A $\frac{c\sqrt{2meV}}{h}$ **B** $\frac{h}{\sqrt{2meV}}$ **C** $\frac{hc}{eV}$ **D** $\frac{eV}{h}$

Ans: A

Loss in EPE = Gain in KE

$$eV = \frac{p^2}{2m}$$

$$p = \sqrt{2meV}$$

$$\lambda = \frac{h}{\sqrt{2meV}}$$

$$f = \frac{c}{\lambda} = \frac{c\sqrt{2meV}}{h}$$

29 The beta spectrum for ^{12}B decay is as shown below